

CS4234: Optimization Algorithms

@Wxy2003-xy

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Introduction and Prerequisites

0.1 Prerequisite map

Source	Reusable idea	Role in CS4234
CS3230	Asymptotic analysis, recurrences, and input encoding	State running times in the true input length and distinguish polynomial from pseudo-polynomial algorithms.
CS3230	Loop invariants, induction, exchange arguments, and optimal substructure	Prove feasibility, optimality, approximation ratios, and termination.
CS3230	Greedy algorithms, dynamic programming, graph algorithms, and randomization	Supply exact subroutines and recurring algorithm-design patterns.
CS3230/CS4269	Polynomial-time reductions and NP-completeness	Explain why an exact polynomial-time algorithm may be unlikely and motivate relaxations, approximation, or restricted cases.
CS4269	Propositional semantics, satisfiability, and formula encodings	Turn combinatorial choices into Boolean or 0–1 variables with explicit constraints.

Topics such as compactness, Hilbert systems, first-order logic, and computability remain useful background, but they are not part of this compact optimization prerequisite.

0.2 The optimization viewpoint

An optimization problem separates the description of legal solutions from the criterion used to rank them.

Definition 0.1 (Optimization problem). For an instance I , let $F(I)$ be its feasible set and let $c_I: F(I) \rightarrow \mathbb{R}$ be its objective. A minimization problem asks for

$$\text{OPT}(I) = \min_{x \in F(I)} c_I(x),$$

while a maximization problem replaces min by max. A complete answer may include both an optimal value and a witness x^* attaining that value.

The same problem often appears in three forms:

- **decision:** is there a feasible solution of value at most B (minimization), or at least B (maximization)?
- **search:** find any feasible solution, possibly satisfying a threshold;
- **optimization:** find the best feasible solution.

For an integer-valued objective known to lie in $[L, U]$, an optimization value can often be recovered using $O(\log(U - L + 1))$ calls to the decision problem. This is a polynomial-time reduction when the endpoints have polynomial bit length; the bound and encoding must be stated rather than silently assumed.

A standard model. Many problems can be written as

$$\min\{c^\top x : x \in F\},$$

where x records choices, F records constraints, and c records costs. The domain of x matters: $x \in \{0, 1\}^n$ is discrete, whereas $x \in [0, 1]^n$ is a continuous relaxation. Enlarging F can make a problem easier to solve, but may produce a solution that is not feasible for the original problem.

0.3 Asymptotics, input size, and cost models

For eventually non-negative functions,

$$\begin{aligned}f \in O(g) &\iff \exists C > 0, \exists n_0, \forall n \geq n_0 : f(n) \leq Cg(n), \\f \in \Omega(g) &\iff \exists c > 0, \exists n_0, \forall n \geq n_0 : f(n) \geq cg(n), \\f \in \Theta(g) &\iff f \in O(g) \cap \Omega(g).\end{aligned}$$

Useful limit tests are

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \Rightarrow f \in o(g), \quad 0 < \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} < \infty \Rightarrow f \in \Theta(g).$$

The robust versions use lim sup and lim inf when the ratio oscillates.

Bit complexity. Running time is measured against the length of the encoded input, not merely the numerical value of its parameters. An integer W requires $\Theta(\log(W+1))$ bits in binary. Consequently, a running time such as $O(nW)$ is polynomial in the *value* W but can be exponential in its bit length. Such an algorithm is **pseudo-polynomial**. This distinction is central for knapsack-style dynamic programs and many weakly NP-hard optimization problems.

Cost accounting. For every claimed bound, identify:

1. the input-size parameters and their encodings;
2. the number of iterations, states, or recursive calls;
3. the cost of each operation or oracle call; and
4. any hidden dependence on numerical precision.

A recurrence such as $T(n) = aT(n/b) + f(n)$ is appropriate for recursive decomposition. Iterative optimization algorithms are often clearer as

$$\text{preprocessing cost} + (\text{number of iterations})(\text{cost per iteration}).$$

0.4 Proof obligations for algorithms

The loop-invariant template from CS3230 remains the default proof pattern: *initialization*, *maintenance*, and *termination*. For a recursive algorithm, replace the invariant by induction on the instance size. For optimization, separate the claims that are easy to accidentally conflate.

1. **Feasibility:** the returned solution lies in $F(I)$.
2. **Value guarantee:** its objective satisfies the promised bound.
3. **Optimality or approximation:** compare it with every feasible solution, a lower/upper bound, or an optimal witness.
4. **Termination and complexity:** the procedure finishes within the claimed resources under the stated computational model.

For a non-negative objective and an α -approximation with $\alpha \geq 1$, the usual conventions are

$$\begin{aligned}\text{minimization: } & \text{OPT}(I) \leq \text{ALG}(I) \leq \alpha \text{OPT}(I), \\ \text{maximization: } & \text{OPT}(I)/\alpha \leq \text{ALG}(I) \leq \text{OPT}(I).\end{aligned}$$

Feasibility gives one side of each inequality; the real proof is the other side. A common strategy is to compare the algorithm against a certificate bound $L(I) \leq \text{OPT}(I)$ for minimization or $U(I) \geq \text{OPT}(I)$ for maximization.

Reusable proof patterns.

- **Exchange argument:** transform an optimum so that it agrees with the greedy choice without worsening its value.
- **Optimal substructure:** after fixing a decision, the remainder is an optimal solution to a well-defined subproblem.
- **Charging argument:** charge the cost of the computed solution to parts of an optimum, with bounded charge per part.
- **Relaxation bound:** compare an integral solution with the value of a relaxation whose feasible region contains all integral solutions.

0.5 Algorithm-design tools retained from CS3230

Dynamic programming. A DP proof should state the subproblem, recurrence, base cases, evaluation order, and recovery of a witness. If $D[s]$ denotes the best value for state s , the key equality is justified by partitioning all feasible solutions by their last decision:

$$D[s] = \operatorname{opt}_{a \in A(s)} \{ \text{cost}(s, a) + D[\text{next}(s, a)] \}.$$

The running time is the number of reachable states times the transition cost. State coordinates that range up to a numeric value such as a capacity W are where pseudo-polynomial behaviour usually enters.

Greedy algorithms. A locally best choice is not a proof. Correctness normally requires an exchange argument, a stays-ahead invariant, or a structural property such as a cut property. When no such proof exists, the same greedy rule may still be useful as a heuristic or approximation algorithm, but its guarantee must be analysed separately.

Graph subroutines. Shortest paths, minimum spanning trees, maximum flow, matching, and connectivity recur as exact subroutines and as sources of relaxations. Always record their preconditions: for example, Dijkstra's algorithm requires non-negative edge weights, whereas Bellman–Ford permits negative edges but must detect reachable negative cycles.

0.6 Probability toolkit for randomized optimization

Let $X_1, \dots, X_n$ be random variables and let $X = \sum_i X_i$. The most reusable facts from CS3230 are

$$\begin{aligned} \mathbb{E}[X] &= \sum_i \mathbb{E}[X_i] && \text{(no independence required),} \\ \Pr\left[\bigcup_i A_i\right] &\leq \sum_i \Pr[A_i] && \text{(union bound),} \\ \Pr[X \geq t] &\leq \frac{\mathbb{E}[X]}{t} && \text{for } X \geq 0 \text{ (Markov).} \end{aligned}$$

Indicator variables $X_i = \mathbf{1}[A_i]$ turn counting questions into expectations because $\mathbb{E}[X_i] = \Pr[A_i]$.

For independent Bernoulli variables with $\mu = \mathbb{E}[X]$ and $0 < \delta < 1$, useful Chernoff forms are

$$\Pr[X \leq (1 - \delta)\mu] \leq e^{-\delta^2 \mu / 2}, \quad \Pr[X \geq (1 + \delta)\mu] \leq e^{-\delta^2 \mu / (2 + \delta)}.$$

These concentration inequalities require stronger assumptions than linearity of expectation; independence must be checked rather than silently assumed.

If one independent run fails with probability at most $q < 1$, then t runs all fail with probability at most q^t . This is the amplification principle seen in Freivalds' algorithm. In optimization it also supports randomized rounding: solve a fractional relaxation, interpret fractional coordinates as selection probabilities, and prove objective and feasibility bounds using expectation plus concentration.

0.7 Logic and constraint modelling from CS4269

A propositional formula has *syntax*; a valuation gives it *semantics*. For a valuation v ,

- $v \models \phi$ means that v satisfies ϕ ;
- ϕ is satisfiable if some v satisfies it;
- ϕ is valid if every v satisfies it;
- $\Gamma \models \phi$ means every model of Γ is a model of ϕ ; and
- $\phi \equiv \psi$ means the two formulas have the same models.

The distinction between satisfiability and validity matters: a feasible solution corresponds to *one* satisfying assignment, not to a formula that is true under all assignments.

Constraint-encoding recipe. The graph-colouring, Hamiltonian-path, clique, exact-cover, and scheduling encodings in CS4269 share one pattern:

1. introduce a variable for each atomic choice;
2. write separate constraints for existence, uniqueness, compatibility, and global requirements;
3. conjoin the constraints into one formula;
4. prove a two-way correspondence between models and feasible solutions;
5. prove that the encoding has polynomial size and is computable in polynomial time.

For variables $p_1, \dots, p_k$, “exactly one” is

$$\left(\bigvee_{i=1}^k p_i \right) \wedge \left(\bigwedge_{1 \leq i < j \leq k} (\neg p_i \vee \neg p_j) \right).$$

The first part gives existence; the pairwise clauses give uniqueness.

0.8 From Boolean constraints to binary optimization

Set $x_i = 1$ when proposition p_i is true and $x_i = 0$ otherwise. Common logical constraints then become linear inequalities:

Boolean constraint	0–1 linear form
$p_1 \vee \dots \vee p_k$	$x_1 + \dots + x_k \geq 1$
$p \rightarrow q$	$x_p \leq x_q$
exactly one of $p_1, \dots, p_k$	$x_1 + \dots + x_k = 1$
$z \leftrightarrow (p \wedge q)$	$z \leq x_p, z \leq x_q, \text{ and } z \geq x_p + x_q - 1$
clause $\ell_1 \vee \dots \vee \ell_k$	$\sum_{\ell_i=p_j} x_j + \sum_{\ell_i=\neg p_j} (1 - x_j) \geq 1$

Example: minimum vertex cover. For a graph $G = (V, E)$, let x_v indicate whether vertex v is selected:

$$\min \sum_{v \in V} x_v \quad \text{subject to} \quad x_u + x_v \geq 1 \quad \forall \{u, v\} \in E, \quad x_v \in \{0, 1\}.$$

Each edge constraint is the clause $p_u \vee p_v$. Replacing $x_v \in \{0, 1\}$ by $0 \leq x_v \leq 1$ gives a linear relaxation. Its feasible region is larger, so for this minimization problem its optimum is a lower bound on the integral optimum.

0.9 Complexity classes and reductions

Definition 0.2 (Core classes). P is the class of decision problems solvable in polynomial time. NP is the class whose YES instances have polynomial-size certificates verifiable in polynomial time. A problem B is NP-hard if every problem in NP polynomial-time reduces to B ; it is NP-complete if, in addition, $B \in NP$.

A polynomial-time many-one reduction $A \leq_p B$ is a polynomial-time computable map f such that

$$x \in A \iff f(x) \in B.$$

The direction has an operational meaning: an algorithm for B would yield an algorithm for A . To prove that B is hard, reduce a known hard problem *to* B , not the other way around.

Every reduction proof has three obligations:

1. **construction:** define $f(x)$ completely;
 2. **correctness:** prove both directions of $x \in A \iff f(x) \in B$; and
 3. **complexity:** prove that f runs in polynomial time and its output has polynomial encoded size.
- The SAT encodings from CS4269 are reductions when they satisfy exactly these obligations. Cook–Levin supplies the central starting point: SAT is NP-complete.

Decision versus optimization hardness. NP-completeness formally applies to decision problems. To discuss an optimization problem, define its threshold version. For example, the decision version of minimum vertex cover asks whether a cover of size at most k exists. If an exact optimizer ran in polynomial time, evaluating whether its returned value is at most k would solve this decision problem in polynomial time. Thus NP-hardness of the threshold problem implies NP-hardness of exact optimization.

0.10 A checklist for the rest of CS4234

For each new problem or algorithm, record the following before doing algebra:

1. What is the instance, and what is its encoded length?
2. What are the variables, feasible solutions, and objective direction?
3. Is the task decision, search, or optimization?
4. What invariant, exchange, induction, charging, or relaxation argument proves the guarantee?
5. Is the guarantee exact, approximate, expected, or with high probability?
6. Does the runtime depend polynomially on the bit length, or only on a numerical value?
7. For a reduction, is the direction correct, is the equivalence proved both ways, and is the construction polynomial in encoded size?

1 Formulae and Theorems

1.1 Basic Probability Identities

$$\Pr(A \cup B) \leq \Pr(A) + \Pr(B) \quad (\text{Union bound})$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) \quad (\text{Inclusion-Exclusion, 2 terms})$$

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \quad A \perp B \iff \Pr(A \cap B) = \Pr(A) \Pr(B)$$

$$\text{Law of Total Probability: } \Pr(A) = \sum_i \Pr(A \mid E_i) \Pr(E_i), \quad \{E_i\} \text{ partition}$$

$$\text{Bayes: } \Pr(H \mid D) = \frac{\Pr(D \mid H) \Pr(H)}{\Pr(D)}$$

1.2 Indicator Trick & Linearity

For indicator $I_E = \mathbf{1}\{E\}$: $\mathbb{E}[I_E] = \Pr(E)$. For a sum $X = \sum_{i=1}^n I_i$ (*counting something*):

$$\mathbb{E}[X] = \sum_{i=1}^n \Pr(I_i = 1) \quad (\text{no independence needed})$$

If I_i are pairwise independent:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(I_i) = \sum_i p_i(1 - p_i)$$

1.3 Expectation/Variance Identities

$$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b \quad (\text{Linearity})$$

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] \quad (\text{Law of total expectation})$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) \quad (\text{Law of total variance})$$

$$\text{Tail-sum (nonneg. integer } X): \mathbb{E}[X] = \sum_{t \geq 1} \Pr[X \geq t]$$

1.4 Inequalities (Concentration)

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \quad (\text{Markov})$$

$$\Pr(|X - \mu| \geq t) \leq \frac{\text{Var}(X)}{t^2} \quad (\text{Chebyshev})$$

Chernoff/Hoeffding for sums of bounded/0–1 r.v.'s. Let $X = \sum_{i=1}^n X_i$, $X_i \in [0, 1]$ independent, $\mu = \mathbb{E}[X]$.

$$\Pr[X \geq (1 + \delta)\mu] \leq \exp\left(-\frac{\mu \delta^2}{2 + \delta}\right) \quad (\delta > 0)$$

$$\Pr[X \leq (1 - \delta)\mu] \leq \exp\left(-\frac{\mu \delta^2}{2}\right) \quad (0 < \delta < 1)$$

$$\Pr(|X - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{n}\right) \quad (\text{Hoeffding, if } X_i \in [0, 1])$$

Handy simplification for $0 < \delta \leq 1$: $\Pr(|X - \mu| \geq \delta\mu) \leq 2 \exp(-\mu\delta^2/3)$.

1.5 Common Inequalities for Mathematical Proofs

Many mathematical arguments rely on a small set of fundamental inequalities. Below is a unified chain collecting the most frequently used algebraic, geometric, convexity, and analytic inequalities. Each tool is stated in a form that is easy to substitute directly into proofs.

Master Chain of Useful Inequalities.

(1) Basic Algebraic Inequalities

$$a \leq b \Rightarrow a + c \leq b + c, \quad a \leq b, c \geq 0 \Rightarrow ac \leq bc.$$

$$(a - b)^2 \geq 0 \Rightarrow 2ab \leq a^2 + b^2 \Rightarrow |ab| \leq \frac{a^2 + b^2}{2}.$$

(2) AM–GM–HM Hierarchy

For all $a, b > 0$,

$$\frac{a + b}{2} \geq \sqrt{ab} \geq \frac{2}{\frac{1}{a} + \frac{1}{b}}.$$

(3) Cauchy–Schwarz Inequality

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

(4) Triangle Inequality

$$|x + y| \leq |x| + |y|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

(5) Jensen's Inequality (Convex Functions)

If f is convex and $\lambda_i \geq 0$, $\sum \lambda_i = 1$,

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

(6) Bernoulli's Inequality

For $x \geq -1$,

$$(1 + x)^r \geq 1 + rx \quad (r \geq 1 \text{ or } r \leq 0).$$

(7) Weighted AM–GM

If $a_i > 0$, $\alpha_i \geq 0$, $\sum \alpha_i = 1$,

$$a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \leq \alpha_1 a_1 + \alpha_2 a_2 + \cdots + \alpha_n a_n.$$

(8) Young’s Inequality

For $p, q > 1$, $\frac{1}{p} + \frac{1}{q} = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

(9) Hölder’s Inequality

$$\|fg\|_1 \leq \|f\|_p \|g\|_q, \quad p, q > 1, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

(10) Logarithmic Inequalities

$$\ln(1+x) \leq x, \quad 1+x \leq e^x, \quad \ln x \leq x-1.$$

(11) Exponential Bounds

$$1+x \leq e^x \leq 1+x + \frac{x^2}{2}e^{|x|}.$$

(12) Chebyshev’s Sum Inequality

For similarly sorted sequences,

$$\frac{1}{n} \sum_{i=1}^n a_i b_i \geq \left(\frac{1}{n} \sum_{i=1}^n a_i \right) \left(\frac{1}{n} \sum_{i=1}^n b_i \right).$$

(13) Rearrangement Inequality

For sorted $a_1 \leq \cdots \leq a_n$,

$$\sum_{i=1}^n a_i b_i \leq \sum_{i=1}^n a_i b_{\sigma(i)} \leq \sum_{i=1}^n a_i b_i^\downarrow.$$

(14) Markov and Chebyshev Bounds

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}, \quad \Pr(|X - \mu| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

(15) Jensen + Logarithm = AM–GM

$$\ln\left(\frac{a_1 + \cdots + a_n}{n}\right) \geq \frac{1}{n} \sum_{i=1}^n \ln a_i.$$

Notes.

- Inequalities (1)–(4) handle almost all algebraic manipulations in proofs.
- (5)–(8) come from convexity and are widely used in optimization and analysis.
- (9)–(13) generalize inner-product arguments and are essential in analysis.
- (14)–(15) appear frequently in algorithm analysis and probability arguments.

1.6 Common Taylor Series Expansions

In this subsection we list the most frequently used Taylor (and Maclaurin) series expansions. Unless stated otherwise, all expansions are Maclaurin series (expansion at $x = 0$) and hold for sufficiently small $|x|$.

(1) Exponential Function

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Useful bounds:

$$1 + x \leq e^x \leq 1 + x + \frac{x^2}{2} e^{|x|}.$$

(2) Natural Logarithm

For $|x| < 1$,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}.$$

Special asymptotic form:

$$\ln(1+x) = x - \frac{x^2}{2} + O(x^3).$$

(3) Geometric Series

For $|x| < 1$,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots.$$

(4) Binomial Series

For any real α , and $|x| < 1$,

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{6} x^3 + \cdots.$$

(5) Sine Function

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

(6) Cosine Function

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}.$$

(7) Tangent (Derivative at 0 and Series)

Though more rarely needed, the Maclaurin series is:

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots.$$

Radius of convergence: $\pi/2$.

(8) Inverse Tangent

For $|x| \leq 1$,

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}.$$

(9) Hyperbolic Sine and Cosine

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}.$$

$$\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}.$$

(10) Square Root Expansion

For $|x| < 1$,

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \cdots.$$

(11) Reciprocal

For $|x| < 1$,

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots.$$

(12) Useful Asymptotic Forms

$$e^x = 1 + x + O(x^2), \quad \ln(1+x) = x - \frac{x^2}{2} + O(x^3),$$

$$\sin x = x + O(x^3), \quad \cos x = 1 - \frac{x^2}{2} + O(x^4).$$

1.7 Common Distributions (means/vars)

$$\text{Bernoulli}(p) : \mathbb{E}[X] = p, \text{ Var}(X) = p(1-p)$$

$$\text{Binomial}(n, p) : \mathbb{E}[X] = np, \text{ Var}(X) = np(1-p)$$

$$\text{Geometric}(p) \text{ (trials to first success)} : \mathbb{E}[X] = \frac{1}{p}, \text{ Var}(X) = \frac{1-p}{p^2}, \text{ memoryless}$$

$$\text{Poisson}(\lambda) : \mathbb{E}[X] = \lambda, \text{ Var}(X) = \lambda \quad (\text{Binomial} \approx \text{Poisson when } n \rightarrow \infty, p \rightarrow 0, np = \lambda)$$

$$\text{Exponential}(\lambda) : \mathbb{E}[X] = \frac{1}{\lambda}, \text{ Var}(X) = \frac{1}{\lambda^2}, \text{ memoryless}$$

$$\text{Normal } \mathcal{N}(\mu, \sigma^2) : \Pr\left(\frac{X - \mu}{\sigma} \in [a, b]\right) \approx \Phi(b) - \Phi(a) \quad (\text{CLT for sums})$$

1.8 Asymptotic Approximations

$$H_n = \sum_{k=1}^n \frac{1}{k} = \ln n + \gamma + O\left(\frac{1}{n}\right)$$

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad (\text{Stirling}) \quad \Rightarrow \quad \binom{n}{k} \approx \frac{n^n}{k^k (n-k)^{n-k}} \sqrt{\frac{n}{2\pi k(n-k)}}$$

1.9 Classic “Balls-into-Bins” Facts

Throw m balls into n bins uniformly and independently.

$$\mathbb{E}[\text{occupied bins}] = n \left(1 - \left(1 - \frac{1}{n}\right)^m\right) \approx n \left(1 - e^{-m/n}\right)$$

$$\Pr(\text{no collisions with } m \text{ balls}) \approx \exp\left(-\frac{m(m-1)}{2n}\right) \quad (\text{birthday paradox})$$

For $m = n$, the maximum load is $\Theta\left(\frac{\log n}{\log \log n}\right)$ w.h.p.

1.10 Coupon Collector

Collect n distinct coupons, each trial draws one uniformly at random.

$$\mathbb{E}[T] = n H_n \approx n \ln n + \gamma n \quad ; \quad \Pr[T > n(\ln n + c)] \leq e^{-c}$$

1.11 Conditioning & Decomposition

$$\mathbb{E}[X] = \sum_i \mathbb{E}[X | E_i] \Pr(E_i), \quad \text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$$

Wald's identity (independent X_i with $\mathbb{E}[X_i] = \mu$, and stopping time N independent of X_i):

$$\mathbb{E}\left[\sum_{i=1}^N X_i\right] = \mathbb{E}[N] \cdot \mu$$

1.12 Jensen & Convexity

For convex g : $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$. Frequently used with $g(x) = e^{tx}$ (mgf) in Chernoff-style proofs.

1.13 “With High Probability” (w.h.p.)

A statement holds w.h.p. if $\Pr[\text{event}] \geq 1 - n^{-c}$ for some constant $c > 0$. Typical pattern: apply a union bound over n events after a Chernoff/Hoeffding bound on each.

Nitpicks that save points: (i) **Linearity** never needs independence; **variance additivity** does. (ii) State the range/independence assumptions before using Chernoff/Hoeffding. (iii) For hashing/“pairwise independence,” Chernoff may fail; use Chebyshev or limited-independence bounds.

Discrete Math Theorems Frequently Used in Competitive Programming

Focus: primes, modular arithmetic, and base conversion/manipulation. Short statements you can drop into proofs or code comments.

1.14 Divisibility, GCD, and Bézout

- **Bézout's identity.** For $a, b \in \mathbb{Z}$, $\exists x, y \in \mathbb{Z}$ with $ax + by = \gcd(a, b)$.
Use: Extended Euclid; modular inverses when $\gcd(a, m) = 1$.
- **LCM–GCD relation.** $\text{lcm}(a, b) \cdot \gcd(a, b) = |ab|$.
- **Euclidean algorithm.** $\gcd(a, b) = \gcd(b, a \bmod b)$, terminates in $O(\log \min\{a, b\})$ steps.

1.15 Modular Arithmetic & Inverses

- **Mod inverse existence.** $a^{-1} \pmod{m}$ exists $\iff \gcd(a, m) = 1$; compute via Extended Euclid.
- **Fermat's little theorem (FLT).** If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; hence $a^{-1} \equiv a^{p-2} \pmod{p}$.
- **Euler's theorem.** If $\gcd(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$.
 φ is multiplicative and $\varphi(p^k) = p^k - p^{k-1}$.
- **Exponent reduction.** For $\gcd(a, m) = 1$, $a^b \bmod m$ can be computed with $b \bmod \varphi(m)$ (careful with $b = 0$ and lifting when $\gcd(a, m) \neq 1$).
- **Wilson's theorem.** p prime $\iff (p-1)! \equiv -1 \pmod{p}$.

1.16 Chinese Remainder Theorem (CRT)

- **Pairwise coprime moduli.** For $x \equiv a_i \pmod{m_i}$, $i = 1..k$, with pairwise coprime m_i and $M = \prod m_i$, the unique solution modulo M is

$$x \equiv \sum_{i=1}^k a_i M_i \overline{M_i} \pmod{M}, \quad M_i = M/m_i, \quad M_i \overline{M_i} \equiv 1 \pmod{m_i}.$$

- **General (two congruences).** $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$ has a solution $\iff a \equiv b \pmod{g}$ where $g = \gcd(m, n)$.

When solvable, solutions form a class modulo $\text{lcm}(m, n)$; one solution is

$$x \equiv a + m \cdot \left(\frac{b-a}{g} \cdot \overline{m/g} \pmod{n/g} \right) \pmod{\text{lcm}(m, n)}.$$

1.17 Prime Factorization Facts

- **Multiplicativity.** If $\gcd(m, n) = 1$, then $\tau(mn) = \tau(m)\tau(n)$ and $\sigma(mn) = \sigma(m)\sigma(n)$, where

$$\tau(n) = \prod_i (e_i + 1), \quad \sigma(n) = \prod_i \frac{p_i^{e_i+1} - 1}{p_i - 1} \quad \text{for } n = \prod_i p_i^{e_i}.$$

- **Legendre's formula.** $v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor$ (exponent of prime p in $n!$).
- **Lifting The Exponent (LTE), common case.** For odd prime $p \mid (a - b)$ and $n \in \mathbb{N}$,

$$v_p(a^n - b^n) = v_p(a - b) + v_p(n).$$

Variants handle $+$, $p = 2$, and when $p \nmid ab$; useful in gcd/valuation problems.

1.18 Primitive Roots & Orders

- **Existence.** A primitive root modulo m exists iff $m \in \{1, 2, 4, p^k, 2p^k\}$ for odd prime p .
- **Order divides totient.** If $\gcd(a, m) = 1$, the multiplicative order $\text{ord}_m(a)$ divides $\varphi(m)$; $a^k \equiv 1 \pmod{m} \iff \text{ord}_m(a) \mid k$.

1.19 Combinatorics Mod Prime

- **Lucas's theorem.** For prime p and $n = \sum n_i p^i$, $k = \sum k_i p^i$ in base p ,

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}.$$

- **Kummer's theorem.** $v_p\left(\binom{n}{k}\right)$ equals the number of carries when adding k and $n - k$ in base p .

1.20 Base Representation & Digit Properties

- **Horner evaluation.** If $N = (d_\ell d_{\ell-1} \dots d_0)_b$, then

$$N = (((d_\ell b + d_{\ell-1})b + d_{\ell-2}) \dots)b + d_0, \quad N \bmod m \text{ via } r := 0; r = (r \cdot b + d_i) \bmod m.$$

- **Divisibility rules (any base $b \geq 2$).**

$$N \equiv \sum_i d_i \pmod{b-1}, \quad N \equiv \sum_i (-1)^i d_i \pmod{b+1} \quad (\text{LSB } i=0).$$

Use: Quick checks for $(b-1)$ or $(b+1)$ divisors; rolling hash invariants.

- **Digit DP invariant.** Counting numbers by digit constraints in base b typically uses state (pos, tight, carry/sum mod $M, \dots$) with transitions $O(b)$ per digit.
- **Base change.** Value is preserved: $(d)_b = \sum_i d_i b^i = \sum_j e_j c^j = (e)_c$. For large integers in arrays, use chunk base $B = 10^k$ to keep operations in 64-bit.

1.21 Sieve & Multiplicative Precomputation

- **Eratosthenes.** Marks primes $\leq n$ in $O(n \log \log n)$. Linear sieve computes primes and SPF in $O(n)$.
- **Multiplicative DP.** For multiplicative f , precompute $f(p^e)$ on the fly with SPF; $f(nm) = f(n)f(m)$ if $\gcd(n, m) = 1$.

1.22 Miscellaneous Handy Facts

- **CRT-friendly mod split.** For $M = 10^k$, choose $M = M_1 M_2$ with M_1, M_2 coprime (e.g. $2^a 5^b$ separated) and work mod M_1 and M_2 then recombine via CRT.
- **Pigeonhole for cycles.** Sequences defined by $x_{i+1} \equiv f(x_i) \pmod{m}$ enter a cycle by step $\leq m+1$ (useful for detecting periods).
- **Binomial mod composite.** Use prime-power decomposition $m = \prod p_i^{e_i}$, compute $\binom{n}{k} \pmod{p_i^{e_i}}$ with factorials excluding p_i and v_{p_i} via Legendre, then CRT.

1.23 Planar Graph: Definition: Kuratowski's theorem

A finite graph is planar if and only if it does not contain a subgraph that is a subdivision of the complete graph K_5 or the complete bipartite graph $K_{3,3}$

If a graph $G = (V, E)$ is planar, the following hold:

- $|E| \leq 3|V| - 6$
- $f \leq 2|V| - 4$
- ($\nexists$ cycle of size 3) $\rightarrow |E| \leq 2|V| - 4$

Euler's formula

If a graph is connected, planar, and finite:

$$v - e + f = 2$$

Subdivision

A subdivision of a graph is obtained by inserting a vertex on an edge 0 or more times. (a graph is a subdivision of itself)

Wagner's theorem

A finite graph is planar if and only if it does not have K_5 or $K_{3,3}$ as a minor.

Minor

A minor of a graph is obtained by contracting an edge into a vertex 0 or more times.

Both being a minor or and being a subdivision of partial orders on graphs up to isomorphism, i.e. only quasi-ordered (Robertson-Seymour theorem)

Topological minor

Subdivision appear as subgraph

Homeomorphic graph

Containing common subdivision

2 Lecture 1: Approximation Algorithm

2.1 Definition

p -Approximation:

- Minimization:

$$ALG \leq p \cdot OPT \implies \frac{ALG}{OPT} \leq p$$

- Maximization:

$$\frac{OPT}{p} \leq ALG \implies \frac{OPT}{ALG} \leq p$$

2.2 General Approach

1. Relax
2. Decompose
3. Compare
4. Conclude

2.3 Example: Greedy algorithms on 0/1 knapsack

We can relax 0/1 knapsack to fractional knapsack, for which prioritizing items by highest value density is optimal. Applying the same ordering to whole items is a natural first greedy approach, but it can be arbitrarily bad for 0/1 knapsack.

Value-density-first greedy

Algorithm 1: Value-density-first greedy for 0/1 knapsack

Input: Values $v_1, \dots, v_n$, weights $w_1, \dots, w_n$, capacity W

Output: A feasible item set S_D and its total value V_D

$F \leftarrow \{i \in \{1, \dots, n\} : w_i \leq W\};$

Sort the items in F by non-increasing density v_i/w_i ;

$S_D \leftarrow \emptyset, W_D \leftarrow 0, V_D \leftarrow 0;$

foreach $i \in F$ *in density order* **do**

if $W_D + w_i > W$ **then**

break;

$S_D \leftarrow S_D \cup \{i\};$

$W_D \leftarrow W_D + w_i;$

$V_D \leftarrow V_D + v_i;$

return $(S_D, V_D);$

Counter example

Consider

$$v = \{2, W\}$$

$$w = \{1, W\}$$

The value-density-first algorithm takes only the first item and returns value 2, while the optimal solution takes the second item and returns W .

$$\frac{OPT}{ALG} = \frac{W}{2}$$

There is no constant approximation factor, meaning this algorithm can give us arbitrarily bad result.

Identify the gap

From the counter example, we can see that a gap is caused by high-value low density (consequently high weight) items. This suggests another greedy approach: prioritize items by value rather than by density.

Algorithm 2: Value-first greedy for 0/1 knapsack

Input: Values $v_1, \dots, v_n$, weights $w_1, \dots, w_n$, capacity W
Output: A feasible item set S_V and its total value V_V
 $F \leftarrow \{i \in \{1, \dots, n\} : w_i \leq W\};$
Sort the items in F by non-increasing value v_i ;
 $S_V \leftarrow \emptyset, W_V \leftarrow 0, V_V \leftarrow 0;$
foreach $i \in F$ *in value order* **do**
 if $W_V + w_i \leq W$ **then**
 $S_V \leftarrow S_V \cup \{i\};$
 $W_V \leftarrow W_V + w_i;$
 $V_V \leftarrow V_V + v_i;$
return $(S_V, V_V);$

This solves the problem of overlooking a high-value item, but it can still be arbitrarily bad.

Counter example for value-first greedy

Let the capacity be $W > 2$, with one heavy item and W unit items:

$$v = \{2, \underbrace{1, 1, \dots, 1}_{W \text{ items}}\},$$
$$w = \{W, \underbrace{1, 1, \dots, 1}_{W \text{ items}}\}.$$

Value-first greedy takes the heavy item, fills the knapsack, and returns value 2. The optimal solution takes all W unit items and has value W . Therefore,

$$\frac{OPT}{ALG} = \frac{W}{2} \rightarrow \infty.$$

Thus neither density-first nor value-first greedy has a constant approximation factor by itself.

Modified greedy: keep the better candidate

The fractional solution is a whole-item density-greedy prefix plus a fraction of the first item that does not fit. This identifies the two candidates needed for the repair: the density-greedy prefix and the best feasible singleton.

Algorithm 3: Modified two-candidate greedy for 0/1 knapsack

Input: Values $v_1, \dots, v_n$, weights $w_1, \dots, w_n$, capacity W
Output: A feasible item set S and its total value
 $F \leftarrow \{i \in \{1, \dots, n\} : w_i \leq W\};$
if $F = \emptyset$ **then**
 return $(\emptyset, 0);$
Sort the items in F by non-increasing density v_i/w_i ;
 $S_G \leftarrow \emptyset, W_G \leftarrow 0, V_G \leftarrow 0;$
foreach $i \in F$ *in density order* **do**
 if $W_G + w_i > W$ **then**
 break;
 $S_G \leftarrow S_G \cup \{i\};$
 $W_G \leftarrow W_G + w_i;$
 $V_G \leftarrow V_G + v_i;$
 $i_{\max} \leftarrow \arg \max_{i \in F} v_i;$
 $S_M \leftarrow \{i_{\max}\}, V_M \leftarrow v_{i_{\max}};$
if $V_G \geq V_M$ **then**
 return $(S_G, V_G);$
else
 return $(S_M, V_M);$

The value-first greedy solution always contains the best feasible singleton, so it could replace the singleton candidate and would preserve the guarantee. The singleton is used because it is the minimal repair required by the proof.

2.4 Strategy: Pick the best among candidate algorithm results

Suppose our goal is to maximize $g(x)$, where

$$g(x) = f_1(x) + f_2(x) + \cdots + f_n(x)$$

But we only have an optimal solution x_i for each individual function f_i . Consider

$$g'(x) = \max_{j \in [1, n]} \left\{ \sum_{i=1}^n f_i(x_j) \right\}$$

We can show that this will give us a solution at worst n times less the optimal solution.

Proof. Suppose there exists an optimal solution x^* , such that $\forall x, g(x^*) \geq g(x)$ thus we have

$$OPT = \sum_{i=1}^n f_i(x^*)$$

Given that we have the local optimal x_i for each f_i , we also have

$$\forall i \leq n, f_i(x^*) \leq f_i(x_i)$$

$$\begin{aligned} OPT &= \sum_{i=1}^n f_i(x^*) \leq \sum_{i=1}^n f_i(x_i) \leq \sum_{j=1}^n \sum_{i=1}^n f_i(x_j) \leq n \times \max_{j \in [1, n]} \left\{ \sum_{i=1}^n f_i(x_j) \right\} = n \cdot ALG \\ ALG &\geq \frac{OPT}{n} \end{aligned}$$

□

The important constraint of picking candidate algorithms is to make sure that

$$OPT \leq \sum_i A_i$$

In the 0/1 knapsack problem, this guarantee is enforced, as the optimal solution for fractional knapsack

$$G + \alpha v_j$$

where $0 \leq \alpha \leq 1$, and v_j is the partially taken item in the greedy solution to the fractional problem. From this we have a series of inequalities:

$$G + \alpha v_j \leq G + v_j \leq G + v_{\max}$$

And as the fractional version is a relaxed version of the integral version, meaning

$$OPT \leq G + \alpha v_j$$

Our modified algorithm gives us

$$ALG = \max(G, v_{\max}) \geq \frac{G + v_{\max}}{2}$$

Overall

$$OPT \leq 2 \cdot ALG$$

2.5 Example: Subset-sum

Consider the optimization version of the subset sum problem, where given a set S , find the max sum of $S' \subseteq S$ such that $\sum S' \leq B$.

Greedy approach

An obvious greedy solution would be to select the largest numbers first.

```

Input: Set  $S$ , threshold  $B$ 
Output: Subset sum  $S'$ 
 $S' \leftarrow \emptyset$ ;
foreach  $s_i \in S$  in descending order do
    if  $B - \sum S' < s_i$  then
        return  $S'$ ;
    else
         $S' \leftarrow S' \cup \{s_i\}$ ;
         $B \leftarrow B + s_i$ ;
return  $S'$ ;

```

We can construct a counter example simiarly to the highest-value first greedy approach for knap-sack: Consider

$$S = \{100, 51, 51\}, B = 102$$

The greedy algorithm gives us 100, while optimal solution is 102. However in this case the greedy algorithm is already a 2-approximation.

Proof. Let the greedy sum be G , and the largest feasible number is M . Given that we select from the largest, we have $G \geq M$ OR $G = 0$. in the case of $G = 0$, meaning there is no feaible element, thus $OPT = ALG = 0$. Remaining 2 non trivial cases:

case 1: $G < \frac{B}{2}$

We will show that $G < \frac{B}{2} \implies G = OPT$. According to our algorithm, suppose a rejection happens after selection some elements, a partial sum $s \leq G$, due to over capacity, meaning

$$s + a_i > B, \quad s > B - a_i$$

Under our assumption,

$$B - a_i < s \leq G < \frac{B}{2}$$

$$a_i > \frac{B}{2}$$

Suppose the first largest item exist in thr selection,

$$G \geq a_1 \geq a_i > \frac{B}{2}$$

contradition, as we assume $G < \frac{B}{2}$

case 2: $G \geq \frac{B}{2}$

given that $OPT \leq B$,

$$\frac{OPT}{2} \leq \frac{B}{2} < G$$

□

2.6 Heuristic

2.7 Set cover

Given a universe U and a family of candidate sets $\mathcal{S}_U$, each $S \in \mathcal{S}_U$ has a non-negative cost $\text{cost}(S)$. The weighted set cover problem asks for a subfamily $S' \subseteq \mathcal{S}_U$ such that

$$\bigcup_{S \in S'} S = U,$$

while minimizing $\sum_{S \in S'} \text{cost}(S)$.

Greedy score

Let R be the set of elements that remain uncovered. For every set that can still cover at least one new element, define

$$\text{score}(S) = \frac{\text{cost}(S)}{|S \cap R|}.$$

This is the average cost per newly covered element. Greedy repeatedly chooses a set of minimum score.

Input: Universe U , candidate family $\mathcal{S}_U$ with costs

Output: A set cover $S' \subseteq \mathcal{S}_U$

$R \leftarrow U$ and $S' \leftarrow \emptyset$;

while $R \neq \emptyset$ **do**

$A_i \leftarrow \arg \min_{\substack{S \in \mathcal{S}_U \\ S \cap R \neq \emptyset}} \frac{\text{cost}(S)}{|S \cap R|};$
 $S' \leftarrow S' \cup \{A_i\};$
 $R \leftarrow R \setminus A_i;$

return S' ;

We prove

$$\text{ALG} \leq H_n \text{OPT} \leq (1 + \ln n) \text{OPT}, \quad H_n = \sum_{j=1}^n \frac{1}{j}, \quad n = |U|.$$

Proof. Assume the instance is feasible, so $\bigcup_{S \in \mathcal{S}_U} S = U$. The invariant is

$$U \setminus R = \bigcup_{A \in S'} A :$$

the elements removed from R are exactly those covered by the sets already selected. If $R \neq \emptyset$, feasibility guarantees a candidate set with $S \cap R \neq \emptyset$, so every iteration removes at least one element from R . Hence the algorithm terminates with $R = \emptyset$ and S' is a cover.

Blue: express the greedy cost exactly. **Red:** bound prices using OPT. **Purple:** combine the two.

1. Rewrite the greedy cost as element prices

Let R_i be the set of uncovered elements at the beginning of iteration i , and let A_i be the set selected in that iteration. The elements newly covered in this iteration are exactly $A_i \cap R_i$. For each $e \in A_i \cap R_i$, define

$$p(e) = \frac{\text{cost}(A_i)}{|A_i \cap R_i|}.$$

This is an accounting price, not an intrinsic cost of e . Already-covered elements of A_i are not charged again. The prices assigned in iteration i recover exactly the cost of A_i :

$$\sum_{e \in A_i \cap R_i} p(e) = |A_i \cap R_i| \frac{\text{cost}(A_i)}{|A_i \cap R_i|} = \text{cost}(A_i).$$

Every element is newly covered exactly once, so summing over all iterations gives the exact identity

$$\text{ALG} = \sum_{A \in S'} \text{cost}(A) = \sum_{e \in U} p(e).$$

2. Use one optimal set to bound its elements' prices

Let $\mathcal{O} \subseteq \mathcal{S}_U$ be a fixed optimal cover and define

$$\text{OPT} = \sum_{S \in \mathcal{O}} \text{cost}(S).$$

Fix any one set $S^* \in \mathcal{O}$ and let $m = |S^*|$. The set S^* may or may not also be selected by greedy; the proof only needs it to be an available candidate for comparison.

Order its elements by when greedy first covers them:

$$S^* = \{e_1, e_2, \dots, e_m\}.$$

If several elements are first covered in the same iteration, break the tie arbitrarily. Let $j(t)$ be the iteration in which e_t is first covered and define

$$r_t = |S^* \cap R_{j(t)}|.$$

Immediately before that iteration, at least $e_t, e_{t+1}, \dots, e_m$ are still uncovered. Therefore

$$r_t \geq m - t + 1.$$

At iteration $j(t)$, greedy chooses $A_{j(t)}$ with minimum score. Comparing its score with the available candidate S^* gives

$$\begin{aligned} p(e_t) &= \frac{\text{cost}(A_{j(t)})}{|A_{j(t)} \cap R_{j(t)}|} \\ &\leq \frac{\text{cost}(S^*)}{|S^* \cap R_{j(t)}|} = \frac{\text{cost}(S^*)}{r_t} \\ &\leq \frac{\text{cost}(S^*)}{m - t + 1}. \end{aligned}$$

The same inequality handles simultaneous coverage: tied elements are all still in $R_{j(t)}$ before that iteration, making r_t larger and the bound only stronger. Summing over the elements of this one optimal set,

$$\begin{aligned} \sum_{e \in S^*} p(e) &= \sum_{t=1}^m p(e_t) \\ &\leq \text{cost}(S^*) \sum_{t=1}^m \frac{1}{m - t + 1} \\ &= \text{cost}(S^*) H_m \leq \text{cost}(S^*) H_n. \end{aligned}$$

This is a bound on the *accounting prices of elements in S^** ; it is not a separate cost paid by greedy to cover only those elements.

3. Bridge the optimal-set bounds back to ALG

Every $e \in U$ belongs to at least one set in the optimal cover $\mathcal{O}$. Thus the middle sum below counts every $p(e)$ at least once (and may count it more than once when optimal sets overlap):

$$\begin{aligned} \text{ALG} &= \sum_{e \in U} p(e) \\ &\leq \sum_{S^* \in \mathcal{O}} \sum_{e \in S^*} p(e) \\ &\leq \sum_{S^* \in \mathcal{O}} H_n \text{cost}(S^*) \\ &= H_n \text{OPT}. \end{aligned}$$

Therefore greedy weighted set cover is an H_n -approximation, and since $H_n \leq 1 + \ln n$, it is also a $(1 + \ln n)$ -approximation. $\square$

2.8 Vertex Cover as a special case of Set cover

Vertex cover can be seen as a special case of set cover with additional structure. In set cover we need to cover all elements, while in vertex cover we are covering edges. The additional structure is the fact that each edge, the element we want to cover, only exist in exactly 2 "sets", which are the vertices.

Stronger structure may give better approximation

```

Input:  $G = (V, E)$ 
Output:  $n$ 
 $M \leftarrow \emptyset$  while  $E \neq \emptyset$  do
    Choose any remaining  $e \in E$ ;
     $M \leftarrow M \cup \{e\}$ ;
    foreach  $e' \in E, e' \text{ incident to } e$  do
         $E \leftarrow E \setminus \{e'\}$ ;
return  $M$ ;

```

We can show that this simple algorithm yields 2-approximation.

Proof. Feasibility

First we need to show the correctness (feasibility) of this algorithm. Consider otherwise, there exist uncovered edges, meaning both endpoints u, v of the edge do not incident to a matched edge $e \in M$, meaning $\{u, v\}$ is disjoint, and therefore could be added to M , contradicting to the maximality of M .

Guarantee

A Matching lower-bounds every possible vertex cover, because matching edges are disjoint, so a cover needs a distinct chosen endpoint for each one, i.e. at least one endpoint for each match.

$$|M| \leq OPT$$

Our algorithm takes both endpoint, thus

$$ALG = 2 \cdot |M| \leq 2 \cdot OPT$$

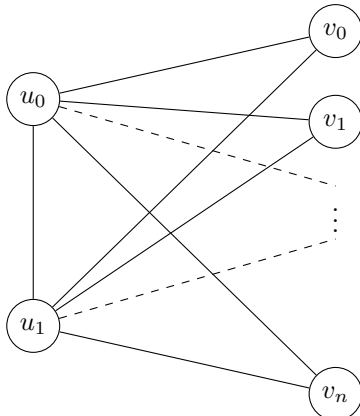
□

Tutorial 1

Q1

Counter example of naive random start greedy algorithm

$$G = (V, E), \quad V = \{u_0, u_1\} \cup \{v_i \mid i \in [0, n]\}, \quad E = \{(u_0, u_1)\} \cup \{(u_0, v_i), (u_1, v_i) \mid i \in [0, n]\}$$



Worst case scenario (u_0, u_1) is never removed, resulting n edges removed instead of 1.

Simple greedy solution gives 3-approx

Similar to vertex cover, here we are looking for "triangle cover". Similar to a maximal matching set, we can find a maximal set of edge-disjoint triangles.

```

Input:  $G = (V, E)$ 
Output: Set  $R$  of edges to remove
 $R \leftarrow \emptyset$ ;
while  $G$  contains a triangle do
    Find any triangle  $(u, v, w)$  in  $G$ ;
     $R \leftarrow R \cup \{(u, v), (v, w), (u, w)\}$ ;
    Remove edges  $(u, v)$ ,  $(v, w)$ , and  $(u, w)$  from  $G$ ;
return  $R$ ;

```

Proof. **Feasibility**

Trivial

Guarantee

In the optimal solution, every disjoint triangle needs to have at least 1 edge removed in OPT , suppose there are k disjoint triangles,

$$k \leq OPT$$

Our algorithm remove all edges of all disjoint triangles,

$$ALG = 3k \leq 3 \cdot OPT$$

□

Patching with heuristic function of each edge

Rank all candidate edges by the number of triangles that can be removed. This can be done in polyno-

```

Input:  $G = (V, E)$ 
Output: Number of edges removed  $n$ 
 $H \leftarrow \emptyset$ ;
triangleCount  $\leftarrow \emptyset$ ;
mial time: foreach  $e = (u, v) \in E$  do
     $H[u] \leftarrow H[u].append(v)$ ;
     $H[v] \leftarrow H[v].append(u)$ ;
    while any edge has triangle count  $> 0$ : access the edge with most triangles, remove, update,
    n++ return  $n$ ;

```

2.9 Fully polynomial-time approximation scheme

Some pseudo polynomial dynamic programming algorithm can be converted to fully polynomial time approximation algorithm by sacrificing some precision to make pseudo polynomial term polynomial to the problem size.

2.9.1 Woeginger FPTAS

Given an algorithm in the form of

```

Input: vectors  $x_1, x_2, \dots, x_n$ , where each vector is made of non negative integers not
    bounded by  $n$ 

```

- $\epsilon :=$ the required approximation ratio
- G : there exists an integer $G \geq 0$
- $1 + \frac{\epsilon}{2Gn}$

Example: Knapsack

First we need to derive given the pseudo polynomial dynamic programming solution that runs in $O(nW)$, we can reduce the number of columns in the DP table, by reducing the resolution. Given an $\epsilon > 0$, max price of any single item $v_{\max}$, define:

- $K = \frac{\epsilon v_{\max}}{n}$
- For each item x_i , define $v'_i = \lfloor \frac{v_i}{K} \rfloor$.

We are defining a rounding down function that convert unbounded variable v_i into v'_i that is bounded by an order of n :

$$f : v_i \mapsto \lfloor \frac{nv_i}{\epsilon v_{\max}} \rfloor$$
$$v'_i \leq \frac{n}{\epsilon}$$

The range of total value becomes

$$R : [0, nv_{\max}] \mapsto [0, n \times \lfloor \frac{n}{\epsilon} \rfloor]$$

Where ϵ is the parameterized approximation factor, and $v_{\max}$ is constant. We can transform the original bound $O(n^2W) = O(n^2v_{\max})$ to $O(n^2 \lfloor \frac{n}{\epsilon} \rfloor)$

First we make the claim that this approximation scheme is feasible (correctness proof)

$$ALG \geq (1 - \epsilon) \cdot OPT$$

Proof. Let $\mathcal{O}$ be the optimal choice set, for any item i , we are rounding down the value, meaning

$$\forall i \in \mathcal{O}, v_i \leq \frac{n}{\epsilon}$$

$$\sum_{i \in \mathcal{O}} v_i - K \times \sum_{i \in \mathcal{O}} v'_i \leq nK$$

□

2.10 Randomized approximation algorithm

2.11 Example: (Minimum) Vertex Cover

2.11.1 Greedy approach

Edge-based greedy selection yields us 2-approximation.

```
Input:  $G = (V, E)$ 
Output:  $V' \subseteq V$ 
 $V' \leftarrow \emptyset;$ 
foreach  $e = (u, v) \in E$  do
    if  $u \in V'$  or  $v \in V'$  then
         $\perp$  continue;
    else
         $b \leftarrow \text{Random}(0, 1);$ 
         $z \leftarrow \emptyset;$ 
        if  $b = 0$  then
             $\perp$   $z \leftarrow u;$ 
        else
             $\perp$   $z \leftarrow v;$ 
         $V' \leftarrow V' \cup \{z\};$ 
         $V \leftarrow V \setminus \{z\};$ 
        foreach  $e' = (u', v') \in E$  do
            if  $z = u'$  or  $z = v'$  then
                 $\perp$   $E \leftarrow E \setminus \{e'\};$ 
return  $V';$ 
```

Proof. Consider the optimal (minimum) vertex cover $V_{OPT} \subseteq V$, and the greedily chosen V_{ALG} . Since our algorithm guarantees that

$$\forall e = (u, v) \in E, u \in V_{ALG} \vee v \in V_{ALG}$$

The same applies to the optimal cover, i.e.

$$\forall e = (u, v) \in E, u \in V_{OPT} \vee v \in V_{OPT}$$

Therefore we have $\forall e = (u, v) \in E$, WLOG, let $|ALG| > |OPT|$. Let A and B denote the set of vertices optimally and not optimally picked,

$$A = ALG \cap OPT, \quad B = ALG \setminus OPT$$

Defined random indicator variable $X_{e=(u,v)}$ such that

$$X_{e=(u,v)} = \begin{cases} 1, & u \text{ or } v \in OPT \\ 0, & u \text{ or } v \notin OPT \end{cases}$$

After running the algorithm, all edges are considered.

$$\mathbb{E}[|A|] = \mathbb{E}\left[\sum_{e \in E} X_{e=(u,v)}\right]$$

By linearity of expectation

$$\mathbb{E}[|A|] = \sum_{e \in E} \mathbb{E}[X_{e=(u,v)}]$$

□

Example: Max Cut

2.12 Example: weighted MAX-SAT

Similarly to vertex cover, a simple random assignment gives us 2-approximation. Given a CNF formula

$$\varphi = \bigwedge_{i=1}^n \bigvee_{j \in [1, m]} l_j$$

with n clauses, and m boolean variables, a random assignment assigns each variable True or False.

Input: CNF φ
Output: Assignment $A \in \{0, 1\}^m$
 $A \leftarrow \emptyset$;
foreach $v \in \varphi.V$ **do**
 $b \leftarrow \text{Random}(0, 1)$;
 $A[v] = b$;
return A ;

Proof. Define independent probability variable x_i for each variable in φ . And indicator $\tau(C_i)$ for each clause in φ , such that

$$\tau(C_i) = \begin{cases} 1, & C_i \text{ is satisfied;} \\ 0, & \text{otherwise} \end{cases}$$

Denote the weight of each clause as w_i , our objective is to find τ to maximize

$$\text{weight}(\tau) = \sum_i w_i \tau(C_i)$$

Unbiased coin: $P(x_i) = \frac{1}{2}$

The probability of clause C_i evaluates to true can be expressed as

$$P(C_i) = 1 - \left(\frac{1}{2}\right)^{|C_i|}$$

We can express the expectation

$$\mathbb{E}[\text{weight}(\tau)] = \sum_i w_i \left(1 - \frac{1}{2^{|C_i|}}\right) \geq \frac{1}{2} \sum_i w_i$$

We know that the optimal is at most the sum of weights of all clauses (when all clauses are satisfied), we have

$$\mathbb{E}[\text{weight}(\tau)] \geq \frac{1}{2} \sum_i w_i \geq \frac{1}{2} \cdot OPT$$

Biased coin

Suppose the coin flip is biased, with a probability of p instead of $\frac{1}{2}$. Let ℓ_i^+ denote the number of positive literals in clause C_i , and ℓ_i^- for negative literals. We can similarly express the probability of one clause evaluates to true as

$$P(C_i) = 1 - (p)^{\ell_i^-} (1-p)^{\ell_i^+}$$

WLOG, let $p \geq \frac{1}{2}$, we have

$$P(C_i) = 1 - p^{\ell_i^-} (1-p)^{\ell_i^+} \geq 1 - p^{\ell_i^-} \cdot p^{\ell_i^+} = 1 - p^{|C_i|}$$

Then we can express the expectation of total weight as

$$\mathbb{E}[\text{weight}(\tau)] = \sum_i w_i [1 - p^{\ell_i^-} (1-p)^{\ell_i^+}] \geq \sum_i w_i (1 - p^{|C_i|})$$

Similar to the unbiased case, we need to construct a lower bound on the probability of each clause to evaluate to true. In the unbiased case, we have $P[C_i] = \frac{1}{2}$ for $|C_i| = 1$, regardless if the singleton clause contains a positive or negative literal. And $\forall |C_j| > |C_i| > 1, (1 - (\frac{1}{2})^{|C_i|}) < (1 - (\frac{1}{2})^{|C_j|})$, which means the probability is monotonously decreasing as the length of the clause.

This is not the case for a biased coin, as we have $p > \frac{1}{2}$, therefore we cannot effectively bound the probability of a clause is satisfiable. For now we patch this by making the following assumption: all singleton contains only positive literals. Now we can construct a non-trivial lower bound

$$q = \min(p, 1 - p^2)$$

Where p represents the case of singleton clauses, and $1 - p^2$ take account of clause of size ≥ 2 , as the monotonicity still holds here. It follows

$$\mathbb{E}[\text{weight}(\tau)] \geq \min(p, 1 - p^2) \sum_i w_i \geq \min(p, 1 - p^2) \cdot OPT$$

Now we have expressed the approximation factor in one variable p , finding the maximum we have

$$\mathbb{E}[\text{weight}(\tau)] \geq \min(p, 1 - p^2) \cdot OPT \approx \frac{\sqrt{5} - 1}{2} \cdot OPT$$

□

2.13 LP relaxation

Similar to propositional logics, we can use integer linear system to model decision and choice based optimization problem. Integer linear programming is NP-hard, but LP in real is not. We may relax a ILP formulation to LP, and round the solutions to integer in a way that the feasibility is preserved.

1. Model problem as an instance of ILP
2. Replace $x_i \in \{0, 1\}$ to $x_i^* \in [0, 1]$, relax to LP
3. solve
4. Rounding

Example: weighted MAX-SAT

Consider the assignment of each individual variable v_i as a binary variable x_i , and the truth value of each clause C_j using a binary variable y_i dependent on the assignment.

- Constraints: $\forall C_i$

$$\sum_{v_j \in C_i} x_j + \sum_{\neg v_k \in C_i} (1 - x_k) \geq y_i$$

- Goal: maximize

Note here we don't need additional clauses to enforce structural constraints like in PL. (may not be the case for other problems, see MAX-DICUT)

Use real solution as expectation to assign integral values

after solving, $\forall x_i$, we assign variable v_i to be true at probability of x_i^*

$$Pr(v_i) = x_i$$

We can express the probability of clause C_i to be true as

$$Pr(C_i) = 1 - \prod_{v_j \in C_i} (1 - x_j^*) \cdot \prod_{\neg v_k \in C_i} x_k^*$$

thus

$$\begin{aligned} \mathbb{E}[\tau(C_i)] &= 1 - \prod_{v_j \in C_i} (1 - x_j^*) \cdot \prod_{\neg v_k \in C_i} x_k^* \\ \sum_{C_i \in \varphi} w_i \cdot \mathbb{E}[\tau(C_i)], \quad OPT &= \sum_{C_i \in \varphi} w_i \cdot y_i^* \end{aligned}$$

Bound Analysis

From our LP formulation

$$\begin{aligned} y_i^* &\leq \sum_{v_j \in C_i} x_j^* + \sum_{\neg v_k \in C_i} (1 - x_k^*) \\ \mathbb{E}[\tau(C_i)] &= 1 - \prod_{v_j \in C_i} (1 - x_j^*) \cdot \prod_{\neg v_k \in C_i} x_k^* \end{aligned}$$

Using AM-GM inequality:

$$\left(\prod_{i=1}^k a_i \right)^{\frac{1}{k}} \leq \frac{1}{k} \sum_{i=1}^k a_i, \quad \prod_{i=1}^k a_i \leq \left(\frac{1}{k} \sum_{i=1}^k a_i \right)^k$$

Use inequality

$$\begin{aligned} \mathbb{E}[\tau(C_i)] &\geq 1 - \left[\frac{1}{|C_i|} \left(\sum_{v_j \in C_i} (1 - x_j^*) + \sum_{\neg v_k \in C_i} x_k^* \right) \right]^{\frac{1}{|C_i|}} \\ &= 1 - \left[1 - \frac{1}{|C_i|} \left(\sum_{v_j \in C_i} x_j^* + \sum_{\neg v_k \in C_i} (1 - x_k^*) \right) \right]^{\frac{1}{|C_i|}} \\ &\geq 1 - \left[1 - \frac{1}{|C_i|} y_i^* \right]^{\frac{1}{|C_i|}} \end{aligned}$$

From standard inequalities

$$1 - x \leq e^{-x}, \quad 1 + x \leq e^x$$

2.14 Inapproximability

3 Linear programming and integer programming

3.1 ILP formulation from finite space FOL

1. Express logical constraints using FOL formula
2. Convert FOL formula to grounded propositional logic formula
3. Binary encoding

Boolean Expression	Linear Constraint
$\bigvee_{i \in I} A_i$	$\sum_{i \in I} A_i \geq 1$
$\bigwedge_{i \in I} A_i$	$\sum_{i \in I} A_i = I $
$\neg \left(\bigvee_{i \in I} A_i \right)$	$\sum_{i \in I} A_i = 0$
$\neg \left(\bigwedge_{i \in I} A_i \right)$	$\sum_{i \in I} A_i \leq I - 1$
$\left(\bigvee_{i \in I} A_i \right) \rightarrow B$	$ I B - \sum_{i \in I} A_i \geq 0$
$\left(\bigwedge_{i \in I} A_i \right) \rightarrow B$	$B - \sum_{i \in I} A_i \geq -(I - 1)$
$A \rightarrow \left(\bigvee_{i \in I} B_i \right)$	$\sum_{i \in I} B_i - A \geq 0$
$A \rightarrow \left(\bigwedge_{i \in I} B_i \right)$	$\sum_{i \in I} B_i - I A \geq 0$

Table 1: Generalized Boolean expressions using binary variables.

Example: Vertex cover

Express the constraints in FOL is simple, define relation $S(v)$ as $v \in VC$

$$\forall (u, v) \in E, S(u) \vee S(v)$$

Define selection indicator variable x_v

$$x_v = \begin{cases} 1, & S(v) \\ 0, & \text{otherwise} \end{cases}$$

Use the encoding, we have

$$\forall (u, v) \in E, \quad x_u + x_v \geq 1$$

The goal becomes to find

$$\min \left(\sum_{v \in V} x_v \right)$$

3.2 Encoding decision variables and facts

Se cover

given a set of elements $U = \{u_1, u_2, \dots, u_n\}$, and a set of sets $\mathcal{S} = \{S_1, S_2, \dots, S_m\}$, assuming $\bigcup_{S \in \mathcal{S}} S = U$, find a set cover to minimize

$$\sum_{S \in \mathcal{S}' \subseteq \mathcal{S}} w_S$$

Decision variable

We will need to use a decision variable to encode

- the coverage of an element: x_i
- the choice of sets: y_j

Define unary relation $S(u_i)$ to denote that u_i is covered, and $R(S_j)$ as set S_j is chosen

$$x_i = \begin{cases} 1, & S(u_i) \\ 0, & \text{otherwise} \end{cases} \quad y_j = \begin{cases} 1, & R(S_j) \\ 0, & \text{otherwise} \end{cases}$$

Beside we can also define more relations to describe the structures, for example $T(u_i, S_j)$ to represent $u_i \in S_j$. Then we can express the constraints as

$$\forall u_i (\exists S_j (T(u_i, S_j) \wedge R(S_j)))$$

3.3 Randomized Rounding

To map LP result in $[0, 1]$ back to $\{0, 1\}$ rounding is needed. The most naive rounding scheme would be to assign x with 1 at probability of x^*

$$\mathbb{E}[x = 1] = x^*$$

Example: MAX-SAT

see above

Example: MAX-DICUT

Sometimes directly using x may not be optimal. For example MAX-DICUT

Let $p_i = \frac{1}{4} + \frac{x_i}{2}$

Recall the LP formulation, the relaxed formulation result is

$$LP^* = \sum_{(i,j) \in A} w_{ij} z_{ij}$$

The probability of an edge being admitted to the dicut is

$$\begin{aligned} P[(i \in U) \wedge j \in W] &= p_i \cdot (1 - p_j) \\ &= \left(\frac{1}{4} + \frac{x_i}{2}\right) \left(1 - \frac{1}{4} - \frac{x_j}{2}\right) \\ &= \left(\frac{1}{4} + \frac{x_i}{2}\right) \left(\frac{1}{4} + \frac{1 - x_j}{2}\right) \end{aligned}$$

From the constraints, which are

$$\begin{aligned} z_{ij} &\leq x_i \\ z_{ij} &\leq 1 - x_j \end{aligned}$$

We have

$$\frac{1}{4} + \frac{x_i}{2} \geq \frac{1}{4} + \frac{z_{ij}}{2}, \quad \frac{1}{4} + \frac{1 - x_j}{2} \geq \frac{1}{4} + \frac{z_{ij}}{2}$$

therefore

$$\begin{aligned} P[(i \in U) \wedge j \in W] &\geq \left(\frac{1}{4} + \frac{z_{ij}}{2}\right)^2 \\ &= \frac{z_{ij}}{2} + \frac{(2z_{ij} - 1)^2}{16} \\ &\geq \frac{z_{ij}}{2} \end{aligned}$$

Define indicator variable X_{ij} to indicate $arc(i, j)$ is in dicut

$$\begin{aligned}
\mathbb{E}[\text{weight}(A)] &= \sum_{(i,j) \in A} w_{ij} \mathbb{E}[X_{ij}] \\
&= \sum_{(i,j) \in A} w_{ij} P[(i \in U) \wedge (j \in W)] \\
&\geq \frac{1}{2} \sum_{(i,j) \in A} w_{ij} z_{ij} \\
&= \frac{1}{2} LP^* \geq \frac{1}{2} \cdot OPT
\end{aligned}$$

Native probability assignemnt

We will see now using $p_i = x_i$ will not give as good approximation.

$$\begin{aligned}
P[(i \in U) \wedge j \in W] &= p_i \cdot (1 - p_j) \\
&= x_i \cdot (1 - x_j) \\
&\geq z_{ij}^2
\end{aligned}$$

thus

$$\begin{aligned}
\mathbb{E}[\text{value}(A)] &= \sum_{(i,j) \in A} w_{ij} \mathbb{E}[X_{ij}] \\
&\geq \sum_{(i,j) \in E} w_{ij} z_{ij}^2
\end{aligned}$$

using Cauchy-scharwz inequality:

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)$$

rearrange by letting $a_{ij} = \sqrt{w_{ij}} z_{ij}$, $b_{ij} = \sqrt{w_{ij}}$

$$\begin{aligned}
\left(\sum_{(i,j) \in E} w_{ij} z_{ij} \right)^2 &\leq \left(\sum_{(i,j) \in E} w_{ij} \right) \left(\sum_{(i,j) \in E} w_{ij} z_{ij}^2 \right) \\
\left(\sum_{(i,j) \in E} w_{ij} z_{ij}^2 \right) &\geq \frac{\left(\sum_{(i,j) \in E} w_{ij} z_{ij} \right)^2}{\sum_{(i,j) \in E} w_{ij}} \\
&= \frac{(LP^*)^2}{\sum_{(i,j) \in E} w_{ij}}
\end{aligned}$$

Let

$$L = LP^*, \quad W = \sum_{(i,j) \in E} w_{ij}$$

$$\begin{aligned}
\mathbb{E}[ALG] &\geq \frac{L^2}{W} = \alpha L \\
\alpha &= \frac{L}{W}
\end{aligned}$$

Fidning optimal linear expression of p in x

Let the probability of assignment p be

$$p_i = a + bx_i$$

Using the same relaxation result, the probability of an edge to be admitted is From $z_{ij} \leq x_i$ and $z_{ij} \leq 1 - x_j$, we have

$$\begin{aligned} P[(i \in U) \wedge (j \in W)] &= p_i(1 - p_j) \\ &= (a + bx_i)(1 - a - bx_j) \\ &= (a + bx_i)((1 - a - b) + b(1 - x_j)) \\ &\geq (a + bz_{ij})(1 - a - b + bz_{ij}). \end{aligned}$$

To force a square term, let $c = 1 - a - b$, such that

$$\begin{aligned} P[(i \in U) \wedge (j \in W)] &\geq (a + bz_{ij})(c + bz_{ij}) \\ &= ac + b(a + c)z_{ij} + b^2 z_{ij}^2 \\ &= ac + b(1 - b)z_{ij} + b^2 z_{ij}^2 \end{aligned}$$

Since how the value of $1 - b$ splits between a and c now only depends on ac , and we have

$$ac \leq \left(\frac{a + c}{2} \right)^2$$

We want to maximize ac , as the difference square term will minus some factor of z_{ij} term: $\gamma \cdot z_{ij}$, and the remaining z_{ij} term will be $[b(1 - b) + \gamma]z_{ij} = \alpha z_{ij}$, where α is the approximation factor which we want to minimize. According to the inequality, ac is maximized when we have

$$a = c \implies a = 1 - a - b, \quad a = \frac{1 - b}{2}$$

Then we substitute c , the original probability bound gives us

$$\begin{aligned} P[(i \in U) \wedge (j \in W)] &\geq (a + bz_{ij})^2 \\ &= 4abz_{ij} + (a - bz_{ij})^2 \\ &\geq 4abz_{ij} \\ \alpha &= 4ab \end{aligned}$$

Substitute a with b ,

$$\begin{aligned} \alpha &= 4ab \\ &= 2b(1 - b) \\ &= -2(b^2 - b + \frac{1}{4}) + \frac{1}{2} \\ &\geq \frac{1}{2} \end{aligned}$$

4 Network flow

4.1 Integer Network flow

4.1.1 Ford–Fulkerson Algorithm

Constraints:

- Capacity constraint:

$$\forall (u, v) \in E, f(u, v) \leq c(u, v)$$

- Conservation constraint:

$$\forall v \in V \setminus \{s, t\}, \sum_{(u, v) \in E} f(u, v) = \sum_{(v, z) \in E} f(v, z)$$

- Goal: maximize $value(f)$, flow leaving s :

$$\sum_{(s, v) \in E} f(s, v)$$

Input: Flow network $G = (V, E)$, capacities $c : E \rightarrow \mathbb{Z}_{\geq 0}$, source s , sink t
Output: Maximum flow f
foreach $e \in E$ **do**
 $f(e) \leftarrow 0$;
Construct residual graph G_f ;
while $\exists s$ - t path P in G_f **do**
 Choose any s - t path P in G_f ;
 $\Delta \leftarrow \min_{(u,v) \in P} c_f(u, v)$;
 foreach $(u, v) \in P$ **do**
 if $(u, v) \in E$ **then**
 $f(u, v) \leftarrow f(u, v) + \Delta$;
 else
 $f(v, u) \leftarrow f(v, u) - \Delta$;
 Update residual graph G_f ;
return f ;

Explanation. Ford–Fulkerson starts with the zero flow and repeatedly searches for an arbitrary augmenting path from the source s to the sink t in the residual graph G_f .

For a current flow f , the residual capacity is defined by

$$c_f(u, v) = \begin{cases} c(u, v) - f(u, v), & (u, v) \in E, \\ f(v, u), & (v, u) \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, the residual graph is

$$G_f = (V, E_f), \quad E_f = \{(u, v) \mid c_f(u, v) > 0\}.$$

If an s - t augmenting path P exists, let

$$\Delta = \min_{(u,v) \in P} c_f(u, v)$$

be the bottleneck residual capacity of the path.

The flow is augmented by Δ along P . For a forward edge $(u, v) \in E$, we increase the flow:

$$f(u, v) \leftarrow f(u, v) + \Delta.$$

For a backward residual edge (u, v) corresponding to an original edge $(v, u) \in E$, we decrease the original flow:

$$f(v, u) \leftarrow f(v, u) - \Delta.$$

After each augmentation, the residual graph is updated. The algorithm terminates when no s - t path exists in the residual graph. By the Max-Flow Min-Cut Theorem, the resulting flow is a maximum flow.

For integer capacities, every residual capacity and every bottleneck value Δ is integral. Therefore, each augmentation increases the flow value by at least 1. Hence Ford–Fulkerson performs at most $|f^*|$ augmentations, where $|f^*|$ is the value of a maximum flow.

If finding an arbitrary augmenting path and updating the flow takes $O(|E|)$ time, the total running time is

$$O(|E| \cdot |f^*|).$$

Thus, Ford–Fulkerson is pseudo-polynomial for integer capacities, since $|f^*|$ may be exponentially large in the number of bits required to encode the capacities.

4.2 $s - t$ cut

Given $G = (V, E)$ and $s, t \in V$ a $s-t$ cut is defined as

$$\{(u, v) \mid u \in A, v \in \bar{A}\}$$

where

$$A \subseteq V, s \in A, t \in \bar{A}$$

Capacity of a cut

Define the capacity of the cut as the sum of the capacity of edges

$$c(A, \bar{A}) = \sum_{(u,v) \in E, u \in A, v \in \bar{A}} c(u, v)$$

Generalized conservation constraint

Consider any $s - t$ cut, $\forall v \in V$

$$f_{out}(v) - f_{in}(v) = \begin{cases} 0, & u \in V \setminus \{s, t\} \\ value(f), & u = s \\ -value(f), & u = t \end{cases}$$

Using the constraint we can define the in/outflow of A in a $s - t$ cut:

$$\begin{aligned} f_{out}(A) &= \sum_{(u,v) \in E, u \in A, v \in \bar{A}} f(u, v) \\ f_{in}(A) &= \sum_{(v,u) \in E, u \in A, v \in \bar{A}} f(u, v) \end{aligned}$$

Lemma 4.1.

$$f_{out}(A) - f_{in}(A) = value(f)$$

Proof.

$$\begin{aligned} f_{out}(A) - f_{in}(A) &= \sum_{u \in A} \sum_{(u,x) \in E} f(u, x) - \sum_{u \in A} \sum_{(x,u) \in E} f(x, u) \\ &= \sum_{u \in A} \left(\sum_{(u,x) \in E} f(u, x) - \sum_{(x,u) \in E} f(x, u) \right) \\ &= \sum_{u \in A} (f_{out}(u) - f_{in}(u)) \\ s \in A &\implies \\ \sum_{u \in A} (f_{out}(u) - f_{in}(u)) &= \sum_{u \in A \setminus \{s\}} 0 + (f_{out}(s) - f_{in}(s)) \\ &= value(f) \end{aligned}$$

□

Corollary 4.2. $f_{out}(A) - f_{in}(A) \leq c(A, \bar{A})$

4.3 Max-flow and min-cut

Max-flow $\leq$ min-cut

We will be proving an equally strong statement: any valid flow $\leq$ any valid cut

Proof. Consider any arbitrary cut, $(A, \bar{A})$, from lemma 3.1, we have

$$\begin{aligned}
\text{value}(f) &= f_{\text{out}}(A) - f_{\text{in}}(A) \\
&\leq f_{\text{out}}(f) \\
&= \sum_{(u,v) \in E, u \in A, v \in \bar{A}} f(u, v) \\
&\leq \sum_{(u,v) \in E, u \in A, v \in \bar{A}} c(u, v)
\end{aligned}$$

Since the partition is arbitrary, any flow through is upper bounded by any cut. $\square$

Max-flow $\geq$ min-cut

Here we will prove that upon FF termination, for partiton $s \in S, t \in T$

$$c(S, T) = \text{value}(f)$$

Proof. Using the FF algorithm, at termination stage, the residual graph has no path from s to t . Suppose otherwise,

If $\exists (u, v) \in E, u \in S, v \in T, f(u, v) < c(u, v)$

Not all edges from S to T are saturated in G , meaning exists residual edges will appear in G_f , contradiction to the termination condition. And by capacity constraint,

$$\forall u \in S, v \in T, f(u, v) = c(u, v)$$

Similarly, we can show that

$$\forall u \in S, v \in T, f(v, u) = 0$$

As otherwise, there exist such a residual edge (u, v) in G_f , t becomes reachable from s , contradicting to the termination condition.

$$\begin{aligned}
\text{value}(f) &= \sum_{(u,v) \in E, u \in S, v \in T} f(u, v) - \sum_{(v,u) \in E, u \in S, v \in T} f(v, u) \\
&= \sum_{(u,v) \in E, u \in S, v \in T} c(u, v) - 0 \\
&= c(S, T)
\end{aligned}$$

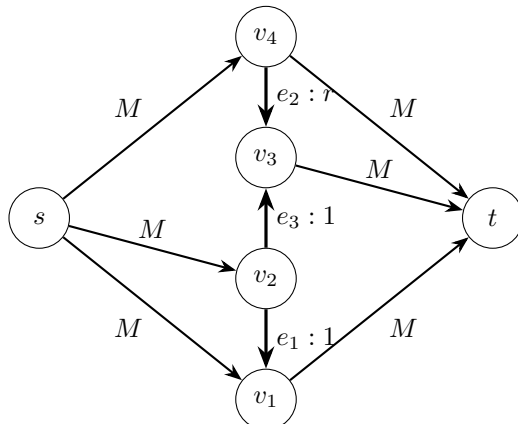
Since upon termination of FF, a valid flow that has the same value as a valid cut are produced, we have

$$\text{min-cut} \leq c(S, T) = \text{value}(f) \leq \text{max-flow}$$

$\square$

4.4 Pathological example of naive FF

FF may not terminate when edge capacity are real, when a particular pathological path choices are made. Consider the following graph:



Pathological augmenting-path sequence. First choose the initial augmenting path

$$P_0 = (s, v_2, v_3, t),$$

which sends one unit of flow and saturates e_3 .

Define the following residual augmenting paths:

$$P_1 = (s, v_4, v_3, v_2, v_1, t),$$

$$P_2 = (s, v_2, v_3, v_4, t),$$

$$P_3 = (s, v_1, v_2, v_3, t).$$

Notice that some edges in these paths are *reverse residual edges* rather than original edges of G . In particular,

$$P_1 : \quad v_3 \rightarrow v_2 \text{ is the reverse residual edge of } e_3,$$

$$P_2 : \quad v_3 \rightarrow v_4 \text{ is the reverse residual edge of } e_2,$$

$$P_3 : \quad v_1 \rightarrow v_2 \text{ is the reverse residual edge of } e_1.$$

After P_0 , repeatedly choose

$$\boxed{P_1, P_2, P_1, P_3}$$

indefinitely. Thus the complete sequence is

$$\boxed{P_0, (P_1, P_2, P_1, P_3), (P_1, P_2, P_1, P_3), (P_1, P_2, P_1, P_3), \dots}$$

If the residual capacities of (e_1, e_2, e_3) are initially

$$(a_n, a_{n+1}, 0),$$

then one four-path cycle gives

$$\begin{aligned} (a_n, a_{n+1}, 0) &\xrightarrow{P_1} (a_{n+2}, 0, a_{n+1}) \\ &\xrightarrow{P_2} (a_{n+2}, a_{n+1}, 0) \\ &\xrightarrow{P_1} (0, a_{n+3}, a_{n+2}) \\ &\xrightarrow{P_3} (a_{n+2}, a_{n+3}, 0). \end{aligned}$$

Here

$$a_{n+2} = a_n - a_{n+1}, \quad a_0 = 1, \quad a_1 = r.$$

Since

$$r^2 = 1 - r,$$

we have

$$a_n = r^n > 0$$

for every finite n . Therefore every cycle leaves another positive-capacity augmenting path, so Ford–Fulkerson never terminates.

Lemma 4.3 (Integrality of max-flow). If all edges have integer capacity, there exists a max-flow that is of integer value.

4.5 Edmonds-Karp

Input: Flow network $G = (V, E)$, capacities $c : E \rightarrow \mathbb{Z}_{\geq 0}$, source s , sink t

Output: Maximum flow f

$k \leftarrow \max(E)$;

foreach $e \in E$ **do**

$f(e) \leftarrow 0$;

Construct residual graph G_f ;

while $k \geq 1$ **do**

while $\exists s-t$ path P in G_f , which has capacity $\geq k$ **do**

 Choose any $s-t$ path P in G_f ;

$\Delta \leftarrow \min_{(u,v) \in P} c_f(u, v)$;

foreach $(u, v) \in P$ **do**

if $(u, v) \in E$ **then**

$f(u, v) \leftarrow f(u, v) + \Delta$;

else

$f(v, u) \leftarrow f(v, u) - \Delta$;

 Update residual graph G_f ;

$k \leftarrow \frac{k}{2}$

return f ;

Upon termination of one inner loop, $\nexists$ an $s-t$ path that has capacity $\geq k$, meaning a "cut" formed at this stage,

- $\forall (u, v) \in E, u \in S, v \in T, c(u, v) - f(u, v) < k$
- $\forall (v, u) \in E, u \in S, v \in T, f(v, u) < k$

Consequently,

$$\begin{aligned}
 \text{value}(f) &= f_{\text{out}}(S) - f_{\text{in}}(S) \\
 &= \sum_{(u,v) \in E, u \in S, v \in T} f(u, v) - \sum_{(v,u) \in E, u \in S, v \in T} f(v, u) \\
 &> \sum_{(u,v) \in E, u \in S, v \in T} (c(u, v) - k) - \sum_{(v,u) \in E, u \in S, v \in T} k \\
 &> c(S, T) - |E|k \\
 &> \text{value}(f_{\max}) - |E|k
 \end{aligned}$$

At the end of this iteration, k is halved. thus we have

$$\text{value}(f) > \text{value}(f_{\max}) - 2k|E|$$

According to the inner loop condition, the path selected has capacity at least k , thus the bottleneck capacity $\Delta \geq k$, each inner loop iteration flow is increased by at least k , thus we need at most $2|E|$ iterations we can reach $f_{\max}$. Therefore the overall complexity is

$$O(|E|^2 \cdot \log_2(c))$$

4.6 Real capacity network flow

Dinitz

Input: Flow network $G = (V, E)$, capacities $c : E \rightarrow \mathbb{Z}_{\geq 0}$, source s , sink t
Output: Maximum flow f
 $k \leftarrow \max(E)$;
foreach $e \in E$ **do**
 $f(e) \leftarrow 0$;
Construct residual graph G_f ;
while *Let P be the shortest $s - t$ path in G_f* **do**
 Choose any $s-t$ path P in G_f ;
 $\Delta \leftarrow \min_{(u,v) \in P} c_f(u, v)$;
 foreach $(u, v) \in P$ **do**
 if $(u, v) \in E$ **then**
 $f(u, v) \leftarrow f(u, v) + \Delta$;
 else
 $f(v, u) \leftarrow f(v, u) - \Delta$;
 Update residual graph G_f ;
return f ;
